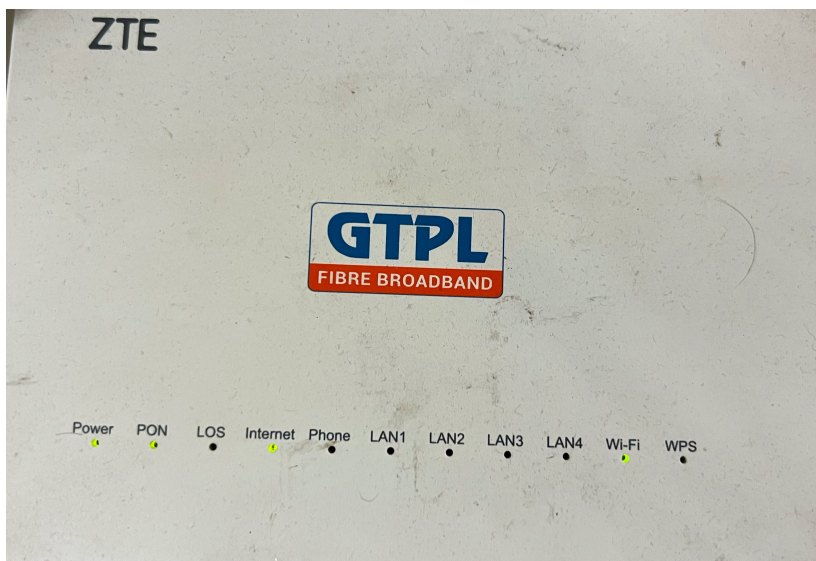
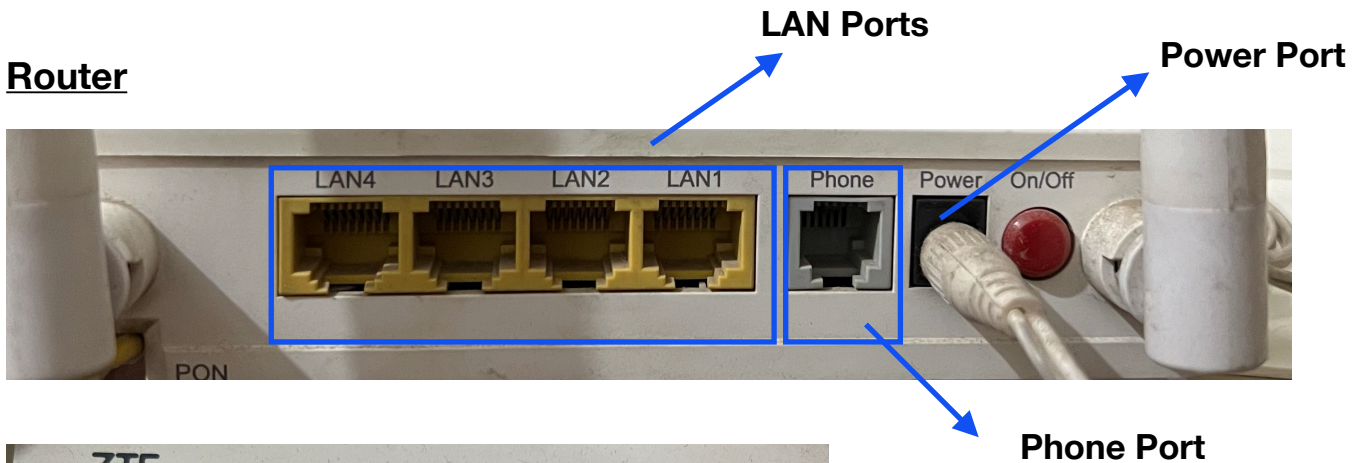


Module 3: A+ Understanding and Maintenance of Networks

DAY 2:-

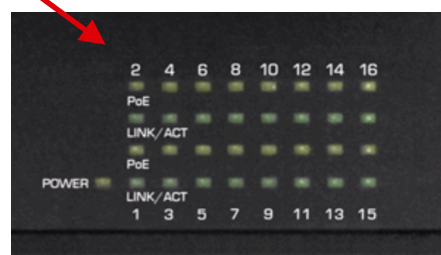
Task 1:

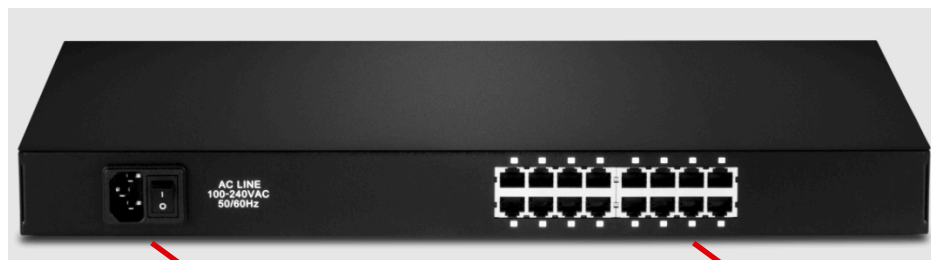
Router



LED Indicators

Switch





LED Indicator

Power Port

LAN Ports

Task 2:

	Router	Switch
Address used for forwarding	Uses IP addresses to route traffic between different networks	Uses MAC addresses to forward traffic between devices on the same network
Networks connected	Connects different networks (e.g., your home LAN to the internet/WAN)	Connects multiple devices within the same LAN
Traffic handling	Makes routing decisions (which path to send data on) and can perform NAT, firewalling, DHCP	Simply forwards frames to the correct port based on a MAC address table — no routing decisions

Task 3:

SAME AS QUES 1.

Task 4:

Setup order: ISP line → Router → Switch → 8 gaming PCs

Why the router connects first:

The router is what connects to the ISP's line and gets the public IP address; it also handles NAT (so all 8 PCs can share one internet connection), DHCP (assigning local IPs automatically), and basic firewalling to protect the internal network. A switch alone can't do any of this — it has no concept of "internet" vs "local network."

Why a switch handles the 8 PCs:

Most home/small-business routers only have 4 LAN ports — not enough for 8 gaming PCs. Instead of buying an 8-port router, you connect one router LAN port to an 8-port (or 16-port, for headroom) switch, and then run an Ethernet cable from the switch to each of the 8 PCs.

Why wired (switch), not Wi-Fi, for the PCs:

For IPL streaming and multiplayer gaming, low latency and stable speed matter a lot — wired Ethernet through a switch avoids Wi-Fi interference, congestion, and lag spikes that would hurt streaming quality and give players input lag during matches.